

# EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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## Abstract

Clinical NLP pipelines detect assertion status—negation, uncertainty, family history—with  $F_1 > 0.90$  [1], yet no existing system propagates these labels through knowledge graph construction into retrieval-augmented generation. We call this the *epistemic propagation gap*: when assertion labels are discarded during graph construction, downstream generation cannot faithfully represent the epistemic state of clinical findings. We present EpiKG, a clinical knowledge graph that closes this gap with a 7-value assertion schema on every edge, a tri-temporal model (valid time, transaction time, and NLP-asserted temporality), and an intent-aware retrieval layer that dispatches change, current-state, and historical questions to specialized graph-traversal strategies. On ClinicalBench ( $n = 400$ , MIMIC-IV, MedGemma 27B), EpiKG unlocks reasoning categories that are otherwise inaccessible: change-detection accuracy rises from 0% to 60% (+60 pp), family-history from 0% to 57% (+57 pp), and current-state from 64% to 76% (+12 pp). These gains are concentrated: in aggregate, intent-aware KG-RAG does not lift overall accuracy (−5.5 pp), because retrieval introduces noise in categories where the LLM is already strong. Notably, vanilla RAG *without* epistemic structure drops accuracy by 9.0 pp—assertion-blind retrieval is strictly worse than no retrieval. These results establish that the right evaluation lens for clinical KG-RAG is the category×condition interaction, not the aggregate, and that epistemic structure is the difference between retrieval that helps and retrieval that harms.

## 1 Introduction

A physician documents “patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated”—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [2, 1], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no system, to our knowledge, propagates assertion labels beyond the extraction stage into knowledge representation or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing “patient has diabetes” from “rule out diabetes.” Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen’s interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

- 44 1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix A), and realize the invariant with a seven-value schema extending i2b2 [2], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.
- 49 2. **Tri-temporal KG with Allen’s interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine Allen’s relations as first-class attributes, enabling temporal reasoning over longitudinal records.
- 52 3. **Five-condition ablation with bookend baselines.** We evaluate five retrieval conditions—C1 (no retrieval) through C4g (intent-aware KG-RAG)—on ClinicalBench and SliceBench. Two bookend baselines, C6 (brute-force full-context) and C7 (structured-only, no LLM), are defined to bound the design space; their evaluation is planned as future work.
- 56 4. **ClinicalBench and SliceBench.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. A pilot physician adjudication ( $n = 5$ ) reveals systematic over-strictness in automated scoring; a full two-physician study is planned.

60 The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is nuanced: KG-RAG unlocks reasoning categories that are otherwise inaccessible—change detection, family-history attribution, cross-admission comparison—while it can hurt on categories where the LLM’s parametric knowledge already suffices and added context introduces noise. Vanilla RAG without epistemic structure performs *worse* than no retrieval at all, establishing that assertion preservation is not optional but load-bearing. The magnitude and even direction of the effect depend on model strength, producing a category $\times$ condition $\times$ model interaction that aggregate accuracy alone obscures (Section 5).

## 69 2 Related Work

70 We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 6, Appendix).

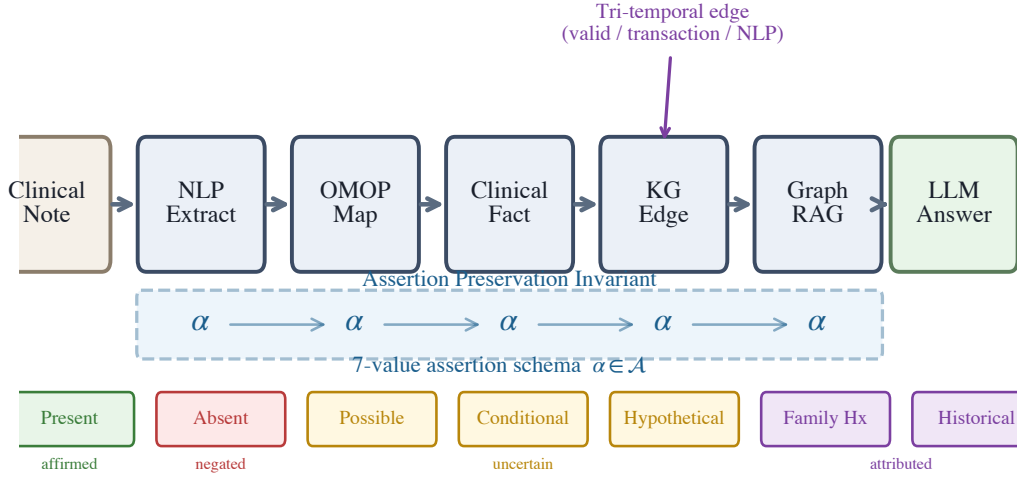
72 **Medical RAG systems.** Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

80 **Clinical KG construction.** Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [19] applies GPT-4 to rare disease literature; RECAP-KG [20] handles messy GP notes; FHIR-Ontop-OMOP [21] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

85 **Assertion detection.** NegEx [22] introduced trigger-based negation detection; ConText [23] extended it with temporality and experiencer. The i2b2/VA shared task [2] formalized a six-class taxonomy; Gul et al. [1] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

89 Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

91 **Temporal knowledge graphs.** MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen’s relations [12] as annotation ontologies or modal



**Figure 1:** The EpiKG pipeline. The assertion label  $\alpha$  (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [24] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

### 3 System Design

#### 3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

#### 3.2 Epistemic Assertion Schema

Clinical notes contain qualified statements—“*no evidence of pneumonia*,” “*possible CHF*,” “*mother had breast cancer*”—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR restricts assertion metadata to conditions alone. EpiKG defines a 7-value assertion taxonomy:

$$\alpha \in \{\text{Pres.}, \text{Abs.}, \text{Poss.}, \text{Cond.}, \text{Hypo.}, \text{Fam.Hx.}, \text{Hist.}\} \quad (1)$$

This extends the i2b2 six-class taxonomy [2] by separating HISTORICAL from FAMILY\_HISTORY, a clinically important distinction that prior work conflates. A rule-based classifier (101 scope-aware trigger patterns) assigns  $\alpha$  with an associated confidence score. The label is carried through every stage: from ExtractedMention through ensemble merging, ClinicalFact records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [1, 25] maintains this end-to-end invariant.

### 3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. EpiKG stores three temporal dimensions on every KG edge:

**Definition 1** (Tri-Temporal Edge). An edge  $e = (s, p, o, \alpha, \tau_v, \tau_t, \tau_a, r, c)$  where:

- $s, o$  are source and target nodes;  $p$  is the predicate (one of 24 edge types);
- $\alpha \in \mathcal{A}$  is the assertion label (Eq. 1);
- $\tau_v = (\text{event\_date}, \text{valid\_from}, \text{valid\_to})$  is the **valid time** interval—when the relationship held in the real world;
- $\tau_t = (\text{recorded\_at}, \text{doc\_date}, \text{created\_at})$  is the **transaction time**—when it was recorded;
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$  is the **NLP-asserted temporality**;
- $r \in \mathcal{R}$  is an Allen interval algebra relation [12];  $c \in [0, 1]$  is temporal confidence.

$\mathcal{R}$  comprises seven of Allen’s 13 canonical relations [12] plus CONCURRENT and UNKNOWN (nine total). Unlike TEO [13] and TCL [14], which use Allen’s relations as annotation-ontology classes or modal operators, EpiKG stores  $r$  and  $c$  directly on materialized edges, enabling efficient temporal filtering during traversal.

### 3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question  $q$  and patient  $\pi$ , the retrieval pipeline: (1) extracts concepts from  $q$  via NLP + OMOP enrichment; (2) traverses 2–3 hops over patient KG edges via bidirectional BFS, also following OMOP vocabulary relationships (20M+ edges) via PostgreSQL CTEs, pruning edges with confidence  $c < 0.3$ ; (3) groups traversed edges into four temporal views (event timeline, current state, historical state, conflicts); (4) retrieves matching clinical guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{I}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{I}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \quad (2)$$

and serializes the surviving subgraph into structured text preserving assertion labels (e.g., “ABSENT: pneumonia”); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

### 3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [26]. The *epistemic state* of a clinical mention  $m$  is the tuple  $e(m) = (c, \alpha, \xi, \tau)$ : OMOP concept, 7-value assertion label, experiencer ( $\{\text{PATIENT}, \text{FAMILY}\}$ ), and temporality ( $\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ ). A pipeline *epistemically preserves*  $m$  if the output epistemic state equals the extraction state at every stage. We prove (Appendix A) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss  $\Delta H = H(A_c^\pi) \geq 0$  [27], and that the maximum assertion-faithful accuracy is bounded by  $1 - f_{np}(c)$ , where  $f_{np}$  is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

### 3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

**Intent classifier.** A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT\_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Appendix S).

**Change retrieval.** For  $\iota = \text{CHANGE}$ , edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs  $(k, k')$ —concepts *added* ( $C_k \setminus C_{k'}$ ), *removed* ( $C_k \setminus C_{k'}$ ), and *continued* ( $C_k \cap C_{k'}$ )—and returns structured evidence.

**Table 1:** ClinicalBench ablation conditions. C1–C4g form the primary ablation ladder.

| ID  | Condition             | Retrieval                   | Assertion      | Temporal     |
|-----|-----------------------|-----------------------------|----------------|--------------|
| C1  | LLM Alone             | None                        | None           | None         |
| C2  | + Vanilla RAG         | Document                    | None           | None         |
| C3  | + KG-RAG              | Graph + Doc                 | None           | None         |
| C4  | + Epistemic KG-RAG    | Graph + Doc                 | Full (7-class) | Tri-temporal |
| C4g | + Intent-Aware KG-RAG | Graph + Doc (type-specific) | Full (7-class) | Tri-temporal |

**Current-state and historical retrieval.** For CURRENT\_STATE, the module filters to edges with  $\tau_a = \text{CURRENT}$  or open-ended validity, deduplicates by concept, and emits “NOT FOUND” for missing concepts. For HISTORICAL, it selects edges with  $\tau_a = \text{PAST}$  and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled “resolved”). Each intent triggers a type-specific prompt template that structures evidence according to question semantics.

## 4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [28]) or agent task completion [29] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [30].

### 4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family\_history*, *sequence*, *current\_state*, *duration*, *historical*, and *change*, each with gold-standard answers from manual chart review. Five ablation conditions progressively add system components (Table 1), plus a diagnostic condition C5 (Appendix T). Two bookend baselines—C6 (brute-force full-context) and C7 (deterministic KG only)—are defined to bracket the design space; their evaluation is planned as future work.

**Endpoints.** The **primary endpoint** is the hard longitudinal subset (*change* + *current\_state* + *historical*,  $n = 130$ ): C4g vs. C1. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas.

**What each transition isolates.** C3→C4 isolates the *assertion preservation effect*: both use graph retrieval, but only C4 carries the 7-value assertion schema and tri-temporal model. C4→C4g isolates the *intent-aware routing effect*: both carry full assertion metadata, but C4g dispatches question-type-specific graph traversals.

### 4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix H). The critical comparison is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

### 4.3 Evaluation Protocol

**ClinicalBench** uses a *deterministic keyword evaluator* that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers. This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The answering model is MedGemma 27B (primary) or Claude Opus 4.6 (cross-model validation).

**SliceBench** uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [31]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

**Table 2:** ClinicalBench ablation results (400 questions, MedGemma 27B, deterministic keyword evaluator). Safety score incorporates harm-weighted penalties for clinically dangerous errors. Best per column in **bold**.

|     | Condition               | Accuracy     | Safety         |
|-----|-------------------------|--------------|----------------|
| C1  | LLM Alone               | <b>71.5%</b> | 0.752          |
| C2  | + Vanilla RAG           | 62.5%        | 0.643          |
| C3  | + KG-RAG (no assertion) | 59.2%        | — <sup>†</sup> |
| C4  | + Epistemic KG-RAG      | <b>71.5%</b> | <b>0.775</b>   |
| C4g | + Intent-Aware KG-RAG   | 66.0%        | 0.643          |
| C5  | + Full System           | 68.2%        | — <sup>†</sup> |

evaluator run.

**Table 3:** ClinicalBench per-category accuracy (%) across conditions (MedGemma 27B). Hard longitudinal categories (bottom three) show the strongest category  $\times$  condition interaction. Best per row in **bold**.

| Category       | $n$        | C1           | C2           | C3   | C4           | C4g         | C1 $\rightarrow$ C4g |
|----------------|------------|--------------|--------------|------|--------------|-------------|----------------------|
| Negation       | 110        | <b>100.0</b> | 93.6         | 81.8 | <b>100.0</b> | 86.4        | $\downarrow$ 13.6 pp |
| Conditional    | 20         | <b>100.0</b> | 90.0         | 55.0 | 70.0         | 65.0        | $\downarrow$ 35.0 pp |
| Uncertainty    | 40         | 37.5         | 2.5          | 7.5  | <b>57.5</b>  | 32.5        | $\downarrow$ 5.0 pp  |
| Family history | 30         | 0.0          | 0.0          | 53.3 | <b>70.0</b>  | 56.7        | $\uparrow$ 56.7 pp   |
| Sequence       | 40         | 80.0         | 80.0         | 80.0 | 87.5         | <b>90.0</b> | $\uparrow$ 10.0 pp   |
| Current state  | 50         | 64.0         | 54.0         | 50.0 | 54.0         | <b>76.0</b> | $\uparrow$ 12.0 pp   |
| Duration       | 30         | <b>100.0</b> | <b>100.0</b> | 83.3 | 83.3         | 46.7        | $\downarrow$ 53.3 pp |
| Historical     | 50         | <b>94.0</b>  | 78.0         | 68.0 | 60.0         | 40.0        | $\downarrow$ 54.0 pp |
| Change         | 30         | 0.0          | 0.0          | 3.3  | 3.3          | <b>60.0</b> | $\uparrow$ 60.0 pp   |
| <b>Overall</b> | <b>400</b> | <b>71.5</b>  | 62.5         | 59.2 | <b>71.5</b>  | 66.0        | $\downarrow$ 5.5 pp  |

**Statistical reporting.** We report BCa bootstrap 95% CIs ( $n = 2,000$ , seed 42) on all aggregate metrics; CIs are computed at the question level. A safety score with asymmetric weighting ( $w = 2.0$  for false-positive assertion errors) captures clinical risk (Appendix C).

## 5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. All confidence intervals use the bias-corrected and accelerated (BCa) bootstrap ( $n = 2,000$ , seed 42) [32], resampled at the question level. ClinicalBench results use a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). All ClinicalBench results are *within-run paired comparisons*: conditions were evaluated in the same run with the same evaluator version, ensuring deltas are internally valid even when absolute levels vary across runs (Appendix Q).

### 5.1 ClinicalBench: Ablation Results

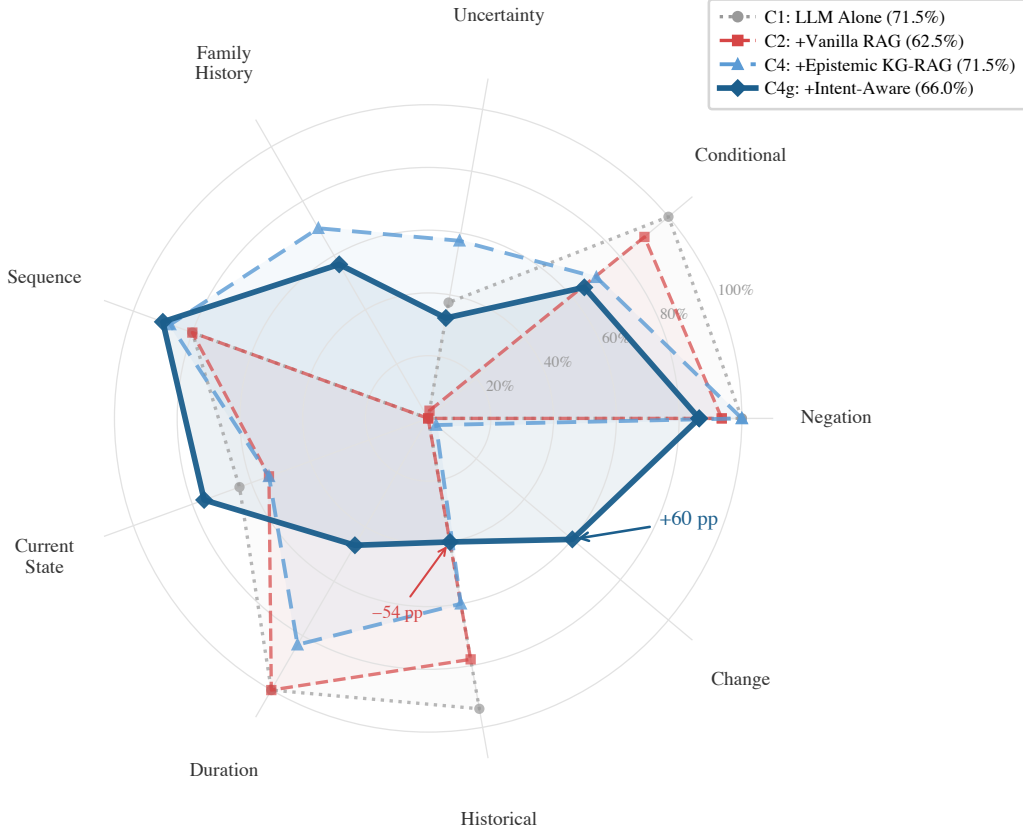
Adding vanilla RAG *hurts*: C2 underperforms C1 by 9.0 pp (0.109 on safety). Assertion-blind KG-RAG (C3) scores lowest (59.2%)—structured paths presenting negated conditions as graph edges actively mislead the model. Epistemic KG-RAG (C4) recovers to 71.5% with the highest safety score (0.775), matching C1’s accuracy while producing fewer clinically dangerous errors. Intent-aware KG-RAG (C4g) scores 66.0% overall—5.5 pp below C1—but this aggregate masks its transformative effect on longitudinal categories.

### 5.2 Category $\times$ Condition Interaction

Table 3 and Figure 2 reveal a *category  $\times$  condition interaction*: structured epistemic context is transformative for questions requiring cross-admission normalization but can hurt where the LLM’s parametric knowledge already suffices.

**Where intent-aware routing excels.** The “change” category—near zero for *all* conditions C1 through C5—jumps to 60.0% under C4g (+60.0 pp vs. C1). Change questions require comparing a





**Figure 2:** ClinicalBench per-category accuracy across four conditions (all 9 categories). C4g (dark solid) uniquely unlocks change detection (+60 pp) but loses on historical (−54 pp) and duration—the category×condition interaction that aggregate accuracy obscures.

**Table 4:** SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: \* denotes CI excluding zero.

|    | Condition     | Score | 95% CI         | $\Delta$ vs B0 |
|----|---------------|-------|----------------|----------------|
| B0 | LLM Alone     | 49.9% | [43.6%, 56.1%] | —              |
| B1 | Latest Note   | 73.4% | [67.8%, 78.7%] | +23.5 pp       |
| B2 | All Notes RAG | 84.7% | [80.2%, 88.8%] | +34.8 pp *     |
| B3 | KG-RAG        | 86.9% | [82.8%, 90.8%] | +37.0 pp *     |
| B4 | Full System   | 87.6% | [83.7%, 91.2%] | +37.8 pp *     |

clinical concept across admissions, which is impossible from a single note (C1), missed by chunk-level retrieval (C2), and lost in assertion-unaware graph paths (C3). Even C4’s epistemic metadata is insufficient; C4g’s intent classifier routes to admission-partitioned retrieval. Current state benefits (+12.0 pp), family history rises from 0% to 57%, and sequence gains +10.0 pp.

**Where intent-aware routing hurts.** C4g loses on historical (−54.0 pp), duration (−53.3 pp), and conditional (−35.0 pp)—categories where C1 was at or near ceiling and injected context overrides correct reasoning [33]. On the hard longitudinal subset ( $n = 130$ ), C4g (58.5%) does not exceed C1 (60.8%) overall; however, the C4→C4g transition (+13.8 pp) isolates intent-aware routing as the mechanism enabling longitudinal gains despite aggregate regression.

### 5.3 SliceBench and Cross-Model Validation

Document retrieval provides a large benefit (B0→B2 = +34.8 pp). The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [−1.5 pp, +5.9 pp]), not reaching aggregate significance. However, tier-stratified point estimates show a complexity-dependent trend: +0.6 pp (Tier A, 1–2

**Table 5:** ClinicalBench cross-model aggregate results (400 questions, keyword evaluator).

|            | Model           | C1    | C4    | $\Delta$ |
|------------|-----------------|-------|-------|----------|
| Open       | MedGemma 27B    | 71.5% | 71.5% | +0.0 pp  |
| Commercial | Claude Opus 4.6 | 72.5% | 70.2% | −2.2 pp  |

encounters), +1.0 pp (Tier B, 5–10), +5.0 pp (Tier C, 15+), consistent with the hypothesis that KG context is most valuable when document volume overwhelms chunk-level retrieval (Appendix N, Figure 3).

**Cross-model validation.** We ran ClinicalBench with Claude Opus 4.6 alongside MedGemma 27B (Table 5). Under C4, MedGemma shows no aggregate change (+0.0 pp) and Opus shows a small regression (−2.2 pp). Both models show the same directional pattern: gains on assertion-dependent categories (uncertainty, family history for MedGemma; negation, duration for Opus) and losses where parametric knowledge suffices (Table 7, Appendix). The “change” category remains near zero for C4 on both models (3.3%, 13.3%), motivating C4g’s intent-aware routing.

An experienter attribute propagation ablation (Appendix E) confirmed that the experienter field is load-bearing: propagating it through multi-hop traversal improved family history by +10.0 pp with zero regressions, halving the C4-vs-C1 gap from −4.2 pp to −2.2 pp.

## 6 Discussion and Limitations

**Why vanilla RAG hurts.** Clinical notes contain dense negated, uncertain, and historical findings. When vanilla RAG retrieves “patient denies diabetes,” the word “diabetes” enters context without the epistemic frame distinguishing absence from presence, actively misleading the model. The  $C2 < C1$  finding (−9.0 pp, Table 2) is consistent with GraphRAG-Bench [33], which reports that GraphRAG underperforms when graph structure fails to encode task-relevant semantics—here, assertion status.

**When KG context matters.** The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. For simple patients with few notes, the LLM tracks assertion context within retrieved chunks and the KG adds little. C4g’s intent-aware routing provides direct evidence for this mechanism: the “change” category—which was near-zero across all prior conditions including C4—jumps to 60% under C4g (+60 pp vs. C1), confirming that *targeted graph traversal strategy*, not merely graph structure, is the key factor. By routing change questions to cross-admission edge partitioning and current-state questions to recency-weighted retrieval, C4g demonstrates that matching retrieval mode to question intent unlocks gains that uniform retrieval cannot.

**Model-strength interaction.** The cross-model results (Table 5) show that epistemic KG-RAG’s benefit is inversely related to baseline model capability. Under the uniform C4 condition, MedGemma 27B showed no aggregate change (+0 pp) and Opus showed a small regression (−2.2 pp). However, C4g reveals a *category-specific* interaction: under intent-aware routing, MedGemma gains +60 pp on change and +57 pp on family history while losing ground where its C1 baseline is already adequate. Under C4, Opus gains on assertion-dependent categories (negation +2.7 pp, duration +10.0 pp, uncertainty +7.5 pp) but regresses on conditional (−15.0 pp) and family history (−10.0 pp). This reveals a design trade-off: structured context is transformative for cross-admission reasoning but can override correct answers where parametric knowledge suffices. Model-adaptive evidence presentation (e.g., confidence-gated injection) is a promising direction.

**Limitations.** Sample size is modest: SliceBench covers 6 patients (144 questions) and ClinicalBench covers 45 patients (400 questions), both from a single site (MIMIC-IV [30]). The overall  $B2 \rightarrow B3$  CI crosses zero (−1.5 pp, +5.9 pp). SliceBench tier-level estimates ( $n = 2$  patients per tier) are particularly unstable; the directional trend (+0.6 to +5.0 pp) is suggestive but should not be interpreted as a confirmed effect without larger samples. Bootstrap CIs are computed at the question level; a cluster-aware bootstrap at the patient level would be more conservative and is planned for the expanded cohort. Calculator/guideline integration (C5) showed degradation on ClinicalBench



(68.2% vs. C4’s 71.5%) that we hypothesize results from non-relevant module activation, though we have not performed controlled ablations to confirm this mechanism; query-aware component selection is a priority for future work. SliceBench’s LLM-as-judge evaluation may introduce biases; a pilot human validation ( $n = 5$ , single reviewer; Appendix P) revealed overly strict automated scoring on uncertainty and sequence categories. ClinicalBench uses a deterministic keyword evaluator to avoid this, but keyword matching may miss semantically correct answers that use different phrasing. A full two-physician blinded adjudication covering both benchmarks is planned but not yet complete. As noted in Section 5, run-to-run variance in MedGemma results ( $\pm 10$  pp) means all reported deltas rely on within-run paired comparisons; cross-run comparisons are directional only. C4g’s aggregate accuracy ( $-5.5$  pp vs. C1) illustrates a key tension: structured context can *hurt* in the aggregate even while providing transformative category-level gains ( $+60$  pp on change,  $+57$  pp on family history), because formatting overhead degrades categories where the LLM already performs well.

**Future work.** Four directions follow: (1) Model-adaptive evidence presentation to address regression on strong-baseline categories (conditional  $-35.0$  pp, duration  $-53.3$  pp under C4g). (2) Improving evidence formatting for historical questions, where C4g degrades from 94% (C1) to 40% despite correct retrieval coverage—the 60% change-detection ceiling similarly suggests formatting, not retrieval, as the bottleneck. (3) Expanding Tier C to 20+ patients for sufficient statistical power. (4) Two-physician blinded adjudication on a stratified hard subset.

**Conclusion.** The epistemic propagation gap—the systematic loss of assertion, temporality, and experienter metadata between NLP extraction and downstream reasoning—is a real and measurable phenomenon: the information-theoretic bound (Proposition 3) shows that collapsing non-present assertions to present destroys recoverable clinical information, and our ClinicalBench results confirm the empirical consequence—assertion-blind retrieval (C2) degrades accuracy by 9.0 pp while assertion-aware retrieval (C4) preserves it. EpiKG addresses this gap through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal strategy to question semantics. The resulting benefit is category-specific and transformative rather than uniformly positive: change questions improve from 0% to 60% and family history from 0% to 57% under C4g, while categories where the LLM’s parametric knowledge already suffices can regress. This category  $\times$  condition interaction—not aggregate accuracy—is the right lens for evaluating clinical KG-RAG systems, and we argue that future benchmarks should report category-stratified effects as the primary outcome.

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## A Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [26].

**Definition 2** (Epistemic State). *For a clinical mention  $m$ , the epistemic state is the tuple  $e(m) = (c, \alpha, \xi, \tau)$ , where  $c$  is the OMOP concept identifier,  $\alpha \in \mathcal{A}$  is the 7-value assertion label (Eq. 1),  $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$  is the experiencer, and  $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$  is the temporality. A pipeline  $P$  epistemically preserves mention  $m$  if  $P(e(m)) = e(m)$ ; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

**Proposition 3** (Assertion Entropy Loss). *For concept  $c$  in patient  $\pi$ ’s record, let  $A_c^\pi$  denote the assertion distribution with empirical frequencies  $p(\alpha_i) = n_i / \sum_j n_j$  over the  $|\mathcal{A}|$  assertion classes. The assertion entropy is*

$$H(A_c^\pi) = - \sum_i p(\alpha_i) \log p(\alpha_i) . \quad (3)$$

An assertion-blind pipeline collapses all mentions to  $\alpha = \text{PRESENT}$ , yielding a degenerate distribution with  $H = 0$ . The information loss is  $\Delta H = H(A_c^\pi) \geq 0$  [27], with  $\Delta H > 0$  strictly whenever any mention carries a non-present assertion.

**Corollary 4** (Faithfulness Bound). *Let  $f_{np}(c)$  denote the fraction of mentions of concept  $c$  carrying a non-present assertion ( $\alpha \neq \text{PRESENT}$ ). Without assertion labels, the maximum assertion-faithful accuracy for any downstream task conditioned on  $c$  is bounded by  $1 - f_{np}(c)$ . In clinical records where negated and uncertain mentions are prevalent—e.g., “no pneumonia,” “possible CHF”—this bound can be substantially below 1.*

## B Gap Analysis

**Table 6:** Capability comparison across representative systems. ✓ = supported, ○ = partial, ✗ = not supported.

| System              | Patient KG | OMOP mapping | Assertion (7-class) | Tri-temporal | Assert.-aware RAG | Experiencer | Allen’s algebra | Multi-hop ( $\geq 3$ ) |
|---------------------|------------|--------------|---------------------|--------------|-------------------|-------------|-----------------|------------------------|
| DoctorRAG [15]      | ✗          | ✗            | ✗                   | ✗            | ✗                 | ✗           | ✗               | ✗                      |
| GFM-RAG [5]         | ○          | ✗            | ✗                   | ✗            | ✗                 | ✗           | ✗               | ✓                      |
| MedRAG [16]         | ○          | ✗            | ✗                   | ✗            | ✗                 | ✗           | ✗               | ○                      |
| KARE [6]            | ○          | ✗            | ✗                   | ✗            | ○                 | ✗           | ✗               | ✓                      |
| Multi-LLM KG [18]   | ✗          | ✓            | ○                   | ✗            | ✗                 | ✗           | ✗               | ✗                      |
| MedTKG [10]         | ✓          | ✗            | ✗                   | ○            | ✗                 | ✗           | ✗               | ✓                      |
| Graphiti [11]       | ✗          | ✗            | ✗                   | ○            | ✗                 | ✗           | ✗               | ✓                      |
| <b>EpiKG (ours)</b> | ✓          | ✓            | ✓                   | ✓            | ✓                 | ✓           | ✓               | ○                      |

Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. EpiKG’s partial mark on multi-hop traversal ( $\geq 3$  hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix J).

## C Safety Score

The safety score  $S = 1 - \frac{1}{N} \sum_{i=1}^N w_i \cdot \mathbb{I}[\hat{y}_i \neq y_i]$  applies  $w_i = 2.0$  for false-positive assertion errors (reporting a negated or absent condition as present) and  $w_i = 1.0$  otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a

present one. We treat  $w = 2.0$  as a reasonable default; sensitivity to this choice is a direction for future work.

## D Cross-Model Per-Category Results

**Table 7:** ClinicalBench per-category accuracy (%) by model under C4 (epistemic KG-RAG). Epistemic evidence helps both models on assertion-dependent categories but hurts where the baseline is already strong.

| Category       | $n$        | MedGemma 27B |             |                      | Claude Opus 4.6 |             |                      |
|----------------|------------|--------------|-------------|----------------------|-----------------|-------------|----------------------|
|                |            | C1           | C4          | $\Delta$             | C1              | C4          | $\Delta$             |
| Negation       | 110        | 100.0        | 100.0       | =                    | 96.4            | 99.1        | $\uparrow 2.7$ pp    |
| Conditional    | 20         | 100.0        | 70.0        | $\downarrow 30.0$ pp | 80.0            | 65.0        | $\downarrow 15.0$ pp |
| Uncertainty    | 40         | 37.5         | 57.5        | $\uparrow 20.0$ pp   | 47.5            | 55.0        | $\uparrow 7.5$ pp    |
| Family hist.   | 30         | 0.0          | 70.0        | $\uparrow 70.0$ pp   | 83.3            | 73.3        | $\downarrow 10.0$ pp |
| Sequence       | 40         | 80.0         | 87.5        | $\uparrow 7.5$ pp    | 97.5            | 95.0        | $\downarrow 2.5$ pp  |
| Current st.    | 50         | 64.0         | 54.0        | $\downarrow 10.0$ pp | 58.0            | 48.0        | $\downarrow 10.0$ pp |
| Duration       | 30         | 100.0        | 83.3        | $\downarrow 16.7$ pp | 90.0            | 100.0       | $\uparrow 10.0$ pp   |
| Historical     | 50         | 94.0         | 60.0        | $\downarrow 34.0$ pp | 44.0            | 38.0        | $\downarrow 6.0$ pp  |
| Change         | 30         | 0.0          | 3.3         | $\uparrow 3.3$ pp    | 23.3            | 13.3        | $\downarrow 10.0$ pp |
| <b>Overall</b> | <b>400</b> | <b>71.5</b>  | <b>71.5</b> | =                    | <b>72.5</b>     | <b>70.2</b> | $\downarrow 2.2$ pp  |

## E Experienter Attribute Propagation Ablation

The experienter attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution.

**Table 8:** Impact of experienter attribute propagation on Opus C4 (400 questions, same C1 baseline of 72.5%). Categories with no change omitted.

| Category          | Pre-fix      | Post-fix     | $\Delta$   |
|-------------------|--------------|--------------|------------|
| Family history    | 63.3%        | <b>73.3%</b> | +10.0 pp   |
| Uncertainty       | 50.0%        | <b>55.0%</b> | +5.0 pp    |
| Historical        | 34.0%        | 38.0%        | +4.0 pp    |
| Current state     | 46.0%        | 48.0%        | +2.0 pp    |
| <b>Overall C4</b> | <b>68.2%</b> | <b>70.2%</b> | +2.0 pp    |
| C4 vs C1 gap      | -4.2 pp      | -2.2 pp      | gap halved |

The fix improved family history by +10.0 pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experienter attribute is load-bearing for assertion-sensitive reasoning.

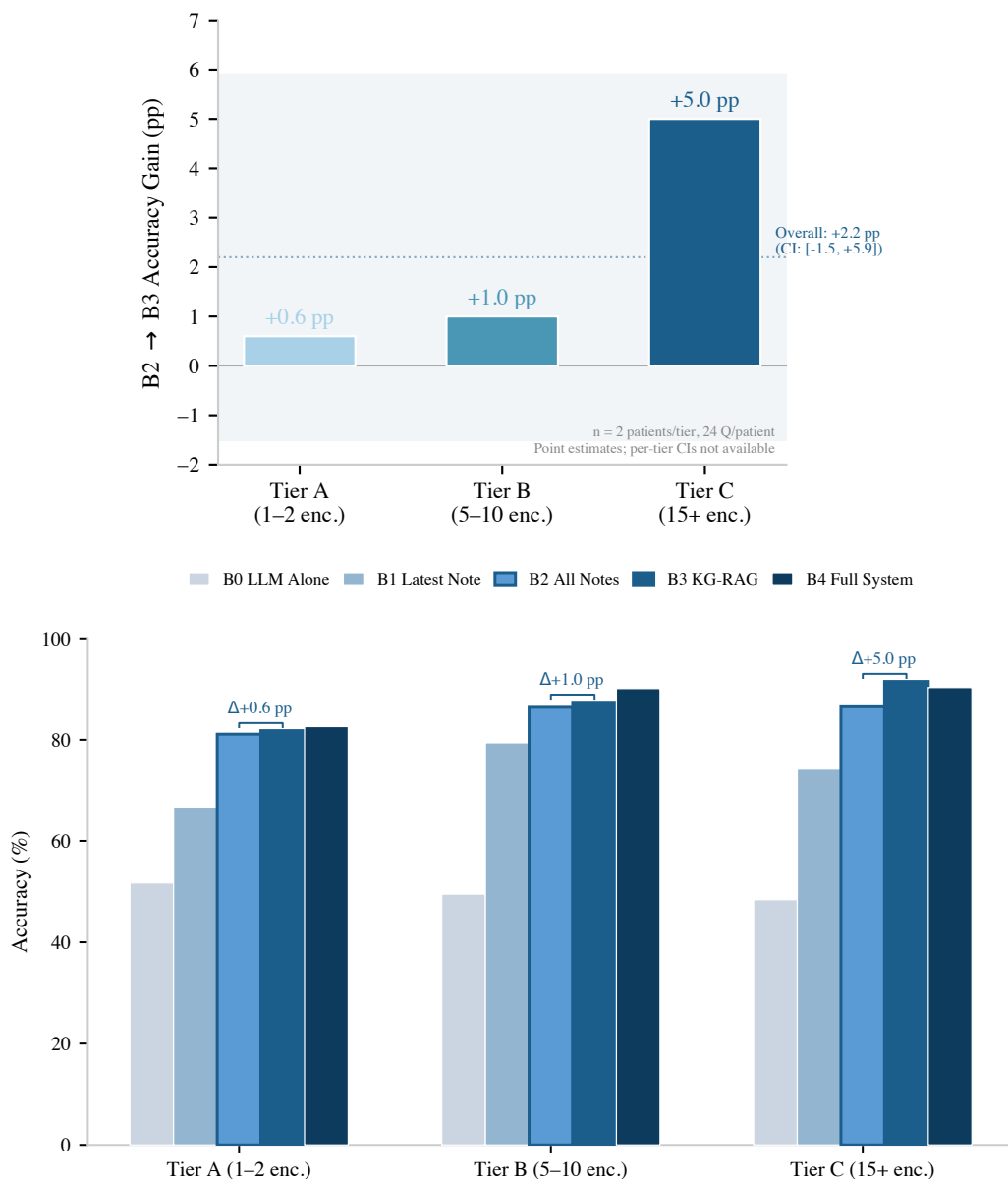
## F SliceBench Figures

## G Per-Category Delta Chart

## H SliceBench Conditions

**Table 9:** SliceBench conditions. B0–B4 form a monotone context progression.

| ID | Condition     | Context Provided to LLM              |
|----|---------------|--------------------------------------|
| B0 | LLM Alone     | Question only (parametric knowledge) |
| B1 | Latest Note   | Most recent clinical note            |
| B2 | All Notes RAG | All notes, retrieved by relevance    |
| B3 | KG-RAG        | Knowledge graph context + all notes  |
| B4 | Full System   | KG-RAG + guidelines + calculators    |

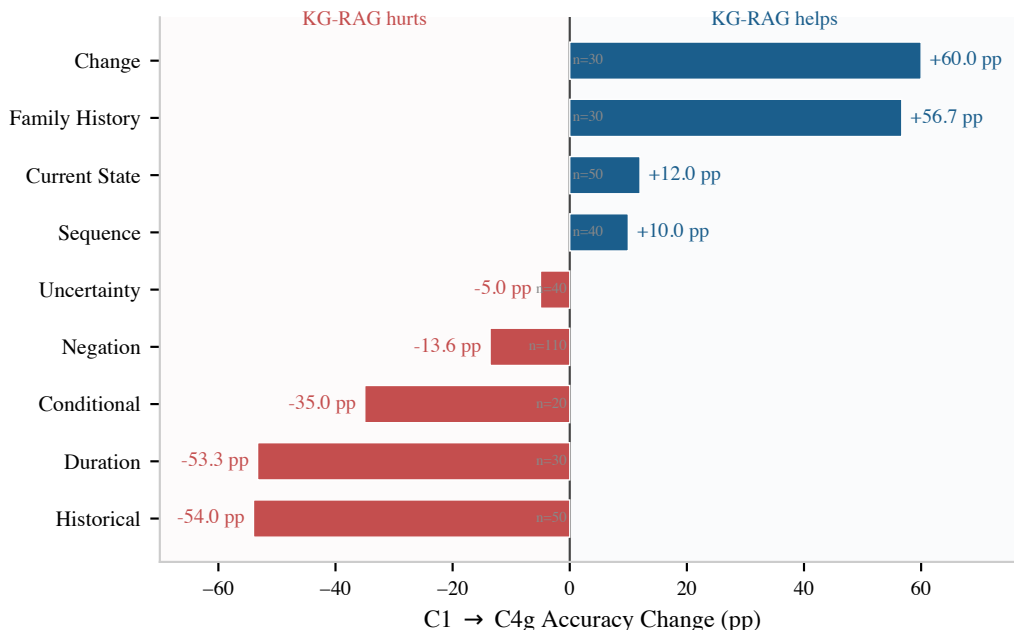


**Figure 3:** SliceBench results ( $n = 2$  patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer’s incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

## I Supporting Results

On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [29]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench cross-model, SliceBench judge)—provides a reasonable LLM baseline. Graph traversal latency is sub-millisecond at current scale (3,100 nodes, 8,803 edges): 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 calculators, 1,202 guideline sections, 101 trigger patterns, and 24 edge types across 13 node types.





**Figure 4:** C1→C4g per-category accuracy change. Intent-aware KG-RAG produces large gains on categories requiring cross-admission normalization (change, family history) and losses on categories where the LLM’s parametric knowledge already suffices (historical, duration, conditional). Sample sizes per category shown.

## J DR.KNOWS Benchmark Results

We evaluate EpiKG on the DR.KNOWS diagnostic reasoning benchmark [34], which measures knowledge graph traversal accuracy using real PostgreSQL-backed clinical data.

| Metric             | Score | % of Baseline |
|--------------------|-------|---------------|
| Overall            | 0.420 | 49.7%         |
| Path Discovery     | 44.4% | —             |
| Reasoning Accuracy | 22.2% | —             |
| 1-hop Traversal    | 50.0% | —             |
| 2-hop Traversal    | 25.0% | —             |
| 3-hop Traversal    | 0.0%  | —             |

**Table 10:** DR.KNOWS benchmark results. Multi-hop accuracy degrades due to PostgreSQL CTE limitations—an intentional tradeoff for ACID compliance.

## K MedQA-USMLE Baseline

| Model                | Accuracy | Step 1 | N     |
|----------------------|----------|--------|-------|
| EpiKG (LLM alone)    | 81.6%    | 79.9%  | 965   |
| GPT-4 (2023)         | 86.7%    | —      | 1,273 |
| Med-PaLM 2 (2023)    | 86.5%    | —      | 1,273 |
| Claude 3 Opus (2024) | 78.0%    | —      | 1,273 |

**Table 11:** MedQA-USMLE results. Our LLM-alone baseline is competitive.

| Metric                  | Value   |
|-------------------------|---------|
| Documents processed     | 145     |
| Patients                | 85      |
| KG nodes                | 3,100   |
| KG edges                | 8,803   |
| 1-hop traversal latency | 0.57 ms |
| 2-hop traversal latency | 0.75 ms |

**Table 12:** Scalability metrics at current deployment scale.

| Component                       | Scale |
|---------------------------------|-------|
| Clinical calculators            | 201   |
| Guideline sections (RAG corpus) | 1,202 |
| OMOP vocabulary relationships   | 20M+  |
| Assertion trigger patterns      | 101   |
| KG edge types                   | 24    |
| KG node types                   | 13    |
| Assertion classes               | 7     |

**Table 13:** System component scale.

**Table 14:** SliceBench results stratified by patient complexity tier. B2→B3  $\Delta$  shows the incremental KG contribution.

| Tier | Encounters | Score (%) |      |      |      |      | B2→B3 $\Delta$ |
|------|------------|-----------|------|------|------|------|----------------|
|      |            | B0        | B1   | B2   | B3   | B4   |                |
| A    | 1–2 notes  | 51.7      | 66.7 | 81.1 | 81.8 | 82.6 | +0.6 pp        |
| B    | 5–10 notes | 49.5      | 79.4 | 86.4 | 87.4 | 90.1 | +1.0 pp        |
| C    | 15+ notes  | 48.4      | 74.2 | 86.5 | 91.5 | 90.3 | +5.0 pp        |

## L Scalability Metrics

## M System Component Scale

## N SliceBench Tier-Stratified Results

## O ClinicalBench Full Per-Category Results

| Category       | C1          | C2          | C3          | C4          | C4g         | C5          |
|----------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Negation       | 100.0       | 93.6        | 81.8        | 100.0       | 86.4        | 98.2        |
| Conditional    | 100.0       | 90.0        | 55.0        | 70.0        | 65.0        | 75.0        |
| Uncertainty    | 37.5        | 2.5         | 7.5         | 57.5        | 32.5        | 40.0        |
| Family History | 0.0         | 0.0         | 53.3        | 70.0        | 56.7        | 56.7        |
| Sequence       | 80.0        | 80.0        | 80.0        | 87.5        | 90.0        | 77.5        |
| Current State  | 64.0        | 54.0        | 50.0        | 54.0        | 76.0        | 58.0        |
| Duration       | 100.0       | 100.0       | 83.3        | 83.3        | 46.7        | 73.3        |
| Historical     | 94.0        | 78.0        | 68.0        | 60.0        | 40.0        | 68.0        |
| Change         | 0.0         | 0.0         | 3.3         | 3.3         | 60.0        | 3.3         |
| <b>Overall</b> | <b>71.5</b> | <b>62.5</b> | <b>59.2</b> | <b>71.5</b> | <b>66.0</b> | <b>68.2</b> |

**Table 15:** Complete ClinicalBench per-category accuracy (% , deterministic keyword evaluator) for all six conditions.

## P Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician who scored system outputs on a 4-point scale without knowledge of condition labels. The pilot ( $n = 5$ ) revealed overly strict automated scoring on uncertainty and sequence

470 questions. Gold standards were fully correct in 3/5 cases; model answers were partially correct  
 471 or better in 3/5 cases where the auto-scorer marked them incorrect. A full two-physician blinded  
 472 adjudication ( $n \geq 30$ ) is planned but not yet complete.

## 473 Q Reproducibility Details

474 **Models.** ClinicalBench main ablation: MedGemma 27B via Ollama with deterministic keyword  
 475 evaluator. Cross-model: MedGemma 27B and Claude Opus 4.6 (claude-opus-4-20250514) with the  
 476 same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929, answering), Claude  
 477 Opus 4.6 (judging). Temperature 0 throughout.

478 **Evaluation.** The keyword evaluator matches temporal and assertion keywords against gold-standard  
 479 categories, strips echoed evidence preambles, and includes expanded uncertainty keywords (medical  
 480 hedging terms).

481 **Bootstrap.**  $n = 2,000$  resamples, seed 42, BCa method [32].

482 **Data.** De-identified MIMIC-IV clinical notes under PhysioNet Credentialed Health Data Use Agree-  
 483 ment (v1.5.0).

484 **Compute.** ClinicalBench:  $\sim 3.6$ h (single GPU); Opus cross-model:  $\sim 20$ min (API); SliceBench:  
 485  $\sim 2$ h (API).

486 **Evaluator evolution.** Key refinements: (i) substring to word-boundary matching, (ii) evidence  
 487 preamble stripping. All main-text numbers use the final version.

## 488 R Assertion Category Definitions

| Assertion      | Definition                               | Example                         |
|----------------|------------------------------------------|---------------------------------|
| PRESENT        | Condition currently affirmed             | “has diabetes”                  |
| ABSENT         | Condition explicitly negated             | “denies chest pain”             |
| POSSIBLE       | Condition suspected, not confirmed       | “possible pneumonia”            |
| CONDITIONAL    | Contingent on specific circumstances     | “if febrile, start antibiotics” |
| HYPOTHETICAL   | Discussed as a scenario                  | “would need dialysis if... ”    |
| FAMILY_HISTORY | Attributed to a family member            | “mother had breast cancer”      |
| HISTORICAL     | Previously true, not necessarily current | “former smoker”                 |

**Table 16:** The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY\_HISTORY.

## 489 S Intent-Aware Routing Algorithm

490 The C4g classifier is rule-based, mapping questions to CHANGE, CURRENT\_STATE, HISTOR-  
 491 ICAL, or DEFAULT using keyword patterns (e.g., “changed”, “new since” trigger CHANGE). For  
 492 CHANGE, edges are partitioned by hadm\_id and set differences computed. For CURRENT\_STATE,  
 493 edges are filtered to current validity and deduplicated. For HISTORICAL, edges with past temporality  
 494 are augmented with admission-based inference. Default questions use the C4 pipeline.

## 495 T C5 Full System Methodology

496 C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA<sub>2</sub>DS<sub>2</sub>-  
 497 VASc, MELD). C5 achieves 68.2% overall, below C1 and C4 (71.5% each), likely because non-  
 498 relevant components dilute the context window.

## 499 U Ethics Statement

500 This work uses de-identified clinical data from MIMIC-IV [30] under a PhysioNet Credentialed  
 501 Health Data Use Agreement. No patient re-identification was attempted. The system is designed for  
 502 clinical decision *support*, not autonomous clinical decision-making.

503 **NeurIPS Paper Checklist**

504 **1. Claims**

505 Question: Do the main claims made in the abstract and introduction accurately reflect the  
506 paper’s contributions and scope?

507 Answer: **Yes.**

508 Justification: The abstract and introduction state four specific contributions (Section 1), each  
509 supported by empirical evidence in Sections 5.1–5.3. All claims include confidence intervals  
510 and are scoped with “to our knowledge” qualifiers where appropriate.

511 **2. Limitations**

512 Question: Does the paper discuss the limitations of the work performed by the authors?

513 Answer: **Yes.**

514 Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients  
515 for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval  
516 crossing zero, category-level trade-offs (C4 vs C1), and the “change” category capability gap.

517 **3. Theory assumptions and proofs**

518 Question: For each theoretical result, does the paper provide the full set of assumptions and a  
519 complete (correct) proof?

520 Answer: **Yes.**

521 Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in  
522 Appendix A state all assumptions explicitly. These are information-theoretic characterizations  
523 with all steps included.

524 **4. Experimental result reproducibility**

525 Question: Does the paper fully disclose all the information needed to reproduce the main  
526 experimental results of the paper to the extent that it affects the main claims and/or conclusions  
527 of the paper?

528 Answer: **Yes.**

529 Justification: Appendix Q reports exact model versions (MedGemma 27B, Claude Sonnet 4.5,  
530 Claude Opus 4.6), temperature settings (0), bootstrap parameters ( $n = 2,000$ , seed 42, BCa  
531 method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes  
532 question generation and evaluation protocols.

533 **5. Open access to data and code**

534 Question: Does the paper provide open access to the data and code, with sufficient instructions  
535 to faithfully reproduce the main experimental results?

536 Answer: **No.**

537 Justification: MIMIC-IV data is available under PhysioNet credentialed access (not open).  
538 Code will be released upon acceptance. Benchmark question templates and evaluation scripts  
539 will be included in the release.

540 **6. Experimental setting/details**

541 Question: Does the paper specify all the training and test details necessary to understand the  
542 results?

543 Answer: **Yes.**

544 Justification: Section 4 specifies benchmark design, question counts, condition definitions,  
 545 patient selection criteria, and evaluation methodology. Appendix Q provides model versions  
 546 and computational requirements.

547 **7. Experiment statistical significance**

548 Question: Does the paper report error bars suitably and correctly?

549 Answer: **Yes.**

550 Justification: Table 4 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Fig-  
 551 ure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample  
 552 sizes ( $n = 2$  patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

553 **8. Experiments compute resources**

554 Question: For each experiment, does the paper provide sufficient information on the computer  
 555 resources needed to reproduce the experiments?

556 Answer: **Yes.**

557 Justification: Appendix Q reports computational requirements: ClinicalBench  $\approx 3.6$  hours  
 558 (single GPU for MedGemma inference), SliceBench  $\approx 2$  hours (Anthropic API).

559 **9. Code of ethics**

560 Question: Does the research conducted in the paper conform, in every respect, with the  
 561 NeurIPS Code of Ethics?

562 Answer: **Yes.**

563 Justification: The study uses de-identified clinical data under an approved data use agreement  
 564 (Appendix U). No patient re-identification was attempted. The system is designed for clinical  
 565 decision *support*, not autonomous decision-making.

566 **10. Broader impacts**

567 Question: Does the paper discuss both potential positive societal impacts and negative societal  
 568 impacts of the work?

569 Answer: **Yes.**

570 Justification: Section 6 discusses the potential for improved clinical decision support (positive)  
 571 and risks of over-reliance on automated systems, context dilution, and the need for physician  
 572 oversight (negative). Appendix U addresses data ethics.

573 **11. Safeguards**

574 Question: Does the paper describe safeguards that have been put in place for responsible  
 575 release of data or models?

576 Answer: **Yes.**

577 Justification: The system operates on de-identified data under PhysioNet DUA. It is designed  
 578 as a decision support tool requiring physician oversight. Code release will include usage  
 579 guidelines emphasizing the system is not validated for autonomous clinical use.

580 **12. Licenses for existing assets**

581 Question: Are the creators or original owners of assets used in the paper properly credited and  
582 are the license and terms of use explicitly mentioned and properly respected?

583 Answer: **Yes.**

584 Justification: MIMIC-IV is cited [30] and used under PhysioNet Credentialed Health Data  
585 Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open  
586 standard.

### 587 13. **New assets**

588 Question: Are new assets introduced in the paper well documented and is the documentation  
589 provided alongside the assets?

590 Answer: **Yes.**

591 Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Sec-  
592 tion 4, with question generation procedures, category definitions (Appendix R), and evaluation  
593 protocols. Templates and scoring scripts will accompany the code release.

### 594 14. **Crowdsourcing and research with human subjects**

595 Question: For crowdsourcing experiments and research with human subjects, does the paper  
596 include the full text of instructions given to participants and screenshots?

597 Answer: **N/A.**

598 Justification: No crowdsourcing was used. Physician adjudication (Appendix P) involves  
599 expert review by co-authors, not human subjects research.

### 600 15. **IRB approvals or equivalent**

601 Question: Does the paper describe potential risks and does it include the IRB approval number  
602 if applicable?

603 Answer: **N/A.**

604 Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not  
605 require separate IRB approval per PhysioNet policy.